

[ISOM2010] INTRODUCTION TO INFORMATION SYSTEMS

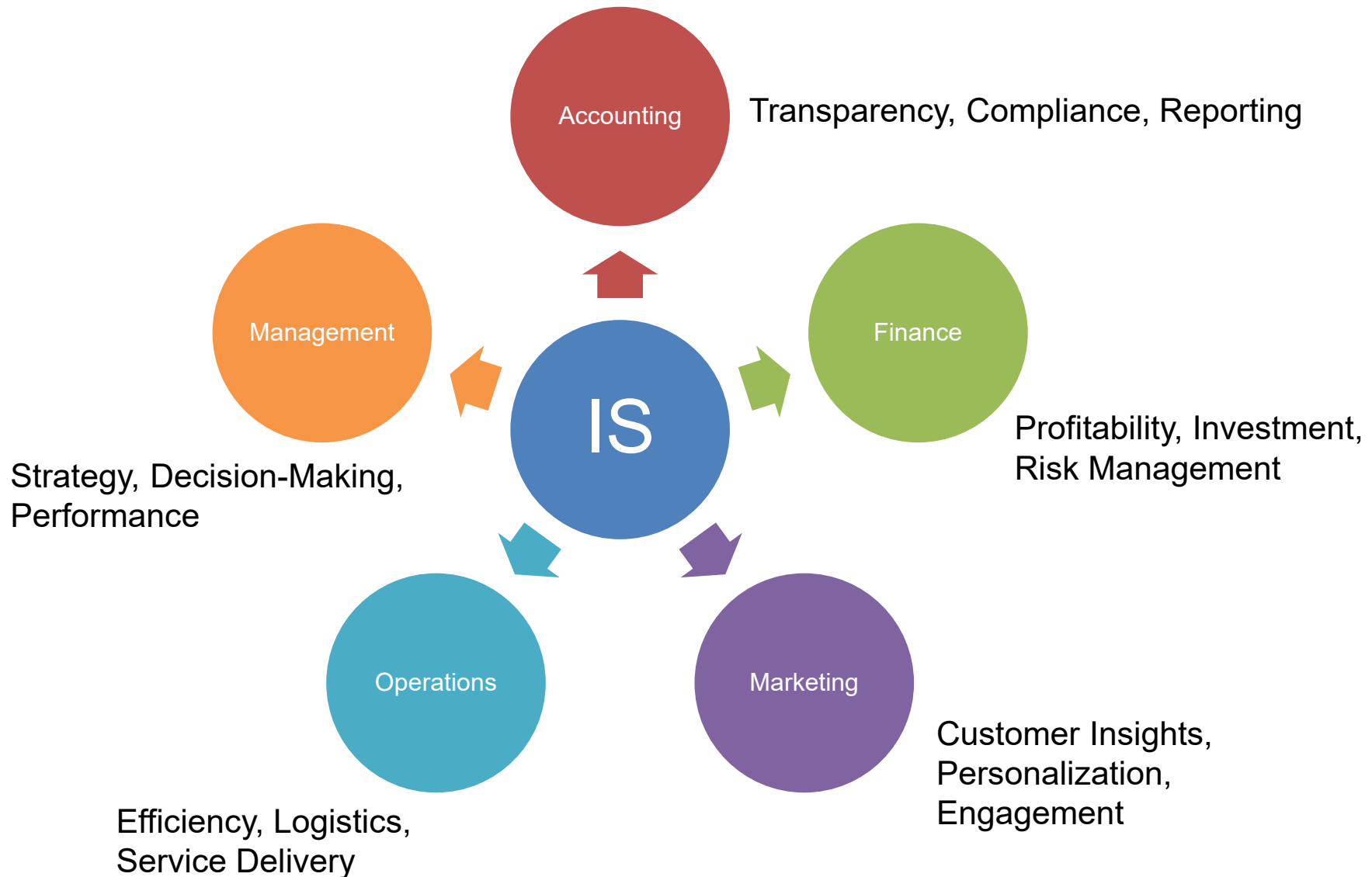
DIGITAL ECONOMY I

: WHAT IS INFORMATION SYSTEMS?

S2

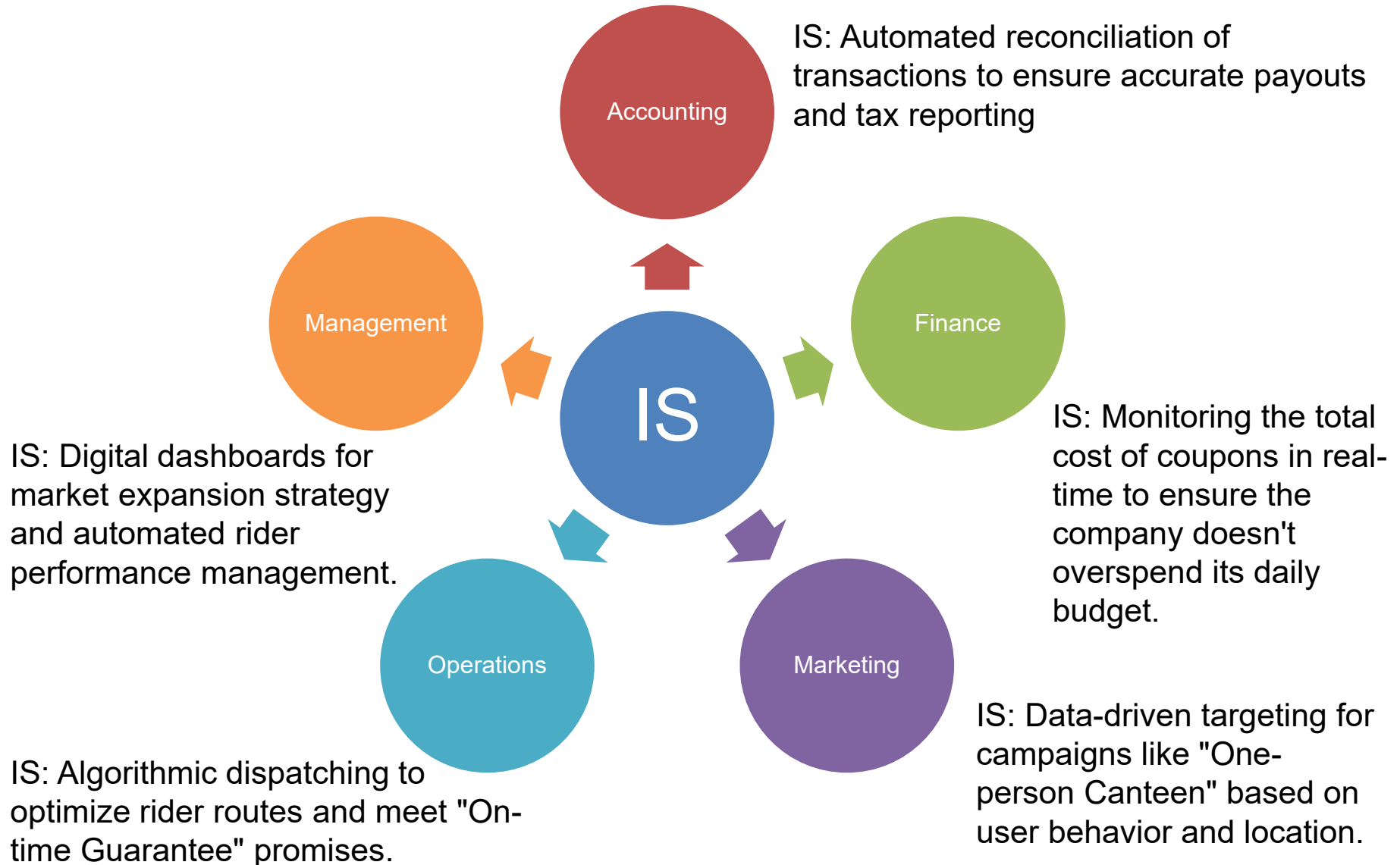
PROF. KIM, YONGSUK

IS: THE ESSENTIAL ENGINE OF THE MODERN ENTERPRISE



IS: THE ESSENTIAL ENGINE OF THE MODERN ENTERPRISE

Delivery App, Keeta's example



WHAT IS AN INFORMATION SYSTEM?

- “Information systems (IS) is the study of complementary networks of hardware and software that people and organizations use to collect, filter, process, create, and distribute data.”^[1]
- “Information systems are combinations of hardware, software, and telecommunications networks that people build and use to collect, create, and distribute useful data, typically in organizational settings.”^[2]
- “Information systems are interrelated components working together to collect, process, store, and disseminate information to support decision making, coordination, control, analysis, and visualization in an organization.”^[3]

1. Wikipedia entry on "Information Systems"

2. Excerpted from *Information Systems Today - Managing in the Digital World*, fourth edition. Prentice-Hall, 2010. ^[4]

3. Excerpted from *Management Information Systems*, twelfth edition, Prentice-Hall, 2012.

THE FIVE-COMPONENT FRAMEWORK OF IS

- These five components are present in every information system – from the most simple to the most complex.

Five-Component Framework



- Why is this important
 - In a business you are part of the information system
 - Most people think of IS as hardware and software only
 - Data, Procedures and People are the more crucial components

THE FIVE-COMPONENT FRAMEWORK OF IS

Five-Component Framework



Hardware (The Infrastructure)

- The "Physical" layer: The tangible devices that process and store data.

Software (The Instructions)

- The "Action" layer: The programs and sets of instructions that tell the hardware what to do.

Data (The Bridge)

- The "Value" layer: The bridge between the computer side (Technical) and the human side (Business).

Procedures (The Business Process)

- The "Human Instructions" layer: The steps and rules that people must follow to make the system work.

People (The Actors)

- The "Decision" layer: The most important and difficult component to manage; the people who interact with the system.

THE FIVE-COMPONENT FRAMEWORK OF IS

Five-Component Framework



Hardware (The Infrastructure)

- e.g., The smartphones used by riders, the merchant tablets in restaurants, and the massive cloud servers that host the platform.

Software (The Instructions)

- e.g., The consumer app you use to order, and the complex dispatching algorithms that calculate the fastest route for a delivery.

Data (The Bridge)

- e.g., Customer order history, real-time GPS coordinates of riders, restaurant menus, and the HK\$10 "One-person Canteen" coupons.

Procedures (The Business Process)

- e.g., The specific steps a rider takes to "confirm delivery" (taking a photo of the food at the door) or the policy for handling a missing item in an order.

People (The Actors)

- e.g., You (the hungry student), the restaurant owner, the delivery rider, and the KeeTa managers analyzing data to plan market expansion.

THE FIVE-COMPONENT FRAMEWORK OF IS

Five-Component Framework



Side	Components	Primary Focus
Technical Side	Hardware & Software	Efficiency: How fast/reliably can we process bits?
The Bridge	Data	Accuracy: What does this information represent?
Business Side	Procedures & People	Strategy: What action should we take to grow?



nature

The AI revolution is running out of data. What can researchers do?

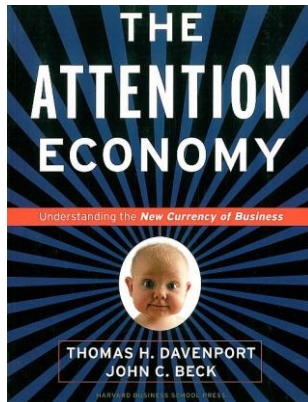
AI developers are rapidly picking the Internet clean to train large language models such as those behind ChatGPT. Here's how they are trying to get around the problem.

▣ *Is this information?*

- 30
- 14.95
- 5
- 25
- 289
- 56389818

DATA, INFORMATION, KNOWLEDGE

➤ *Is this information?*



Price: \$14.95

Inventory: 30

Location: 5 on shelf, 25 in warehouse

Sold: 8391

...

DATA, INFORMATION, KNOWLEDGE

📌 *Is this information?*



ID: 56389818

Remaining balance: HKD\$289

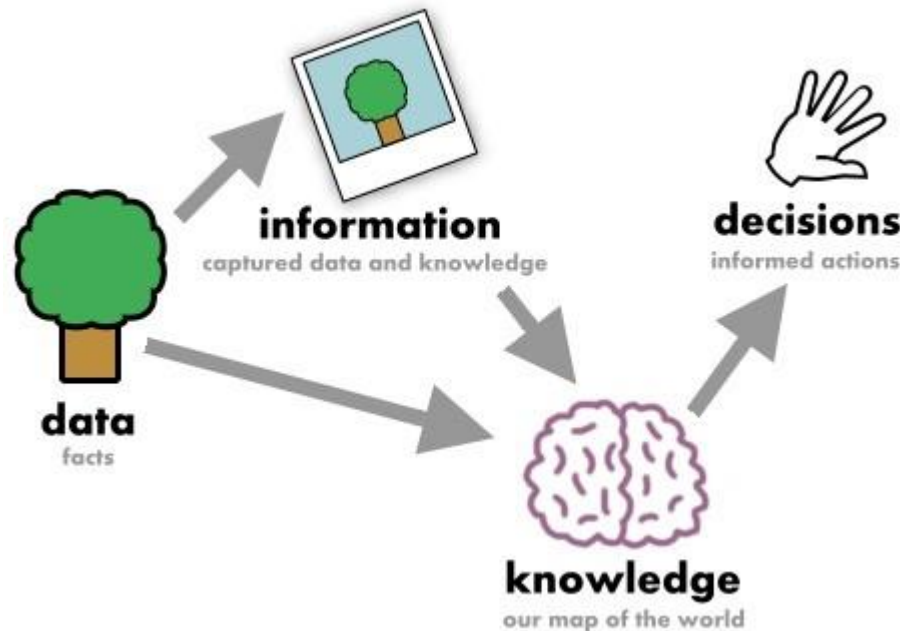
Linked to credit card: yes

Last 10 transactions: ...

DATA, INFORMATION, KNOWLEDGE

Information

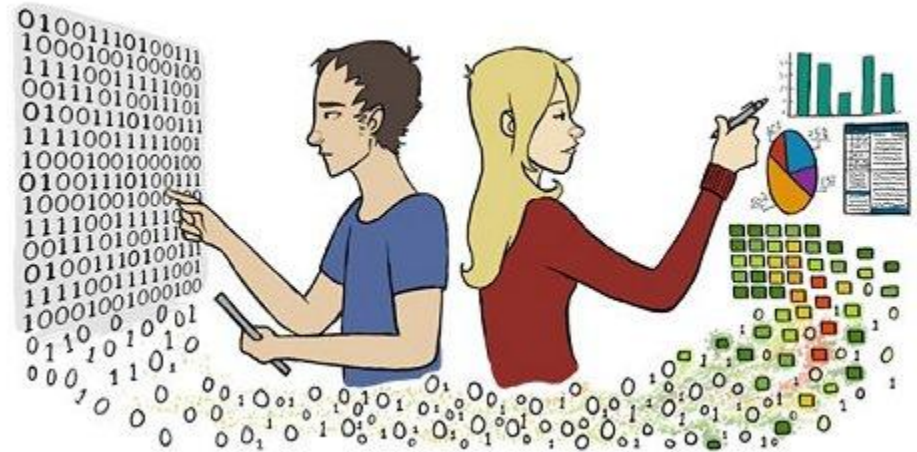
- **Useful, meaningful** data for users (a person, an organization...) in a specific **context**.
- Different information can be generated from the same set of **data** by analyzing it from **different angles** or applying it in **different contexts**.



DATA, INFORMATION, KNOWLEDGE

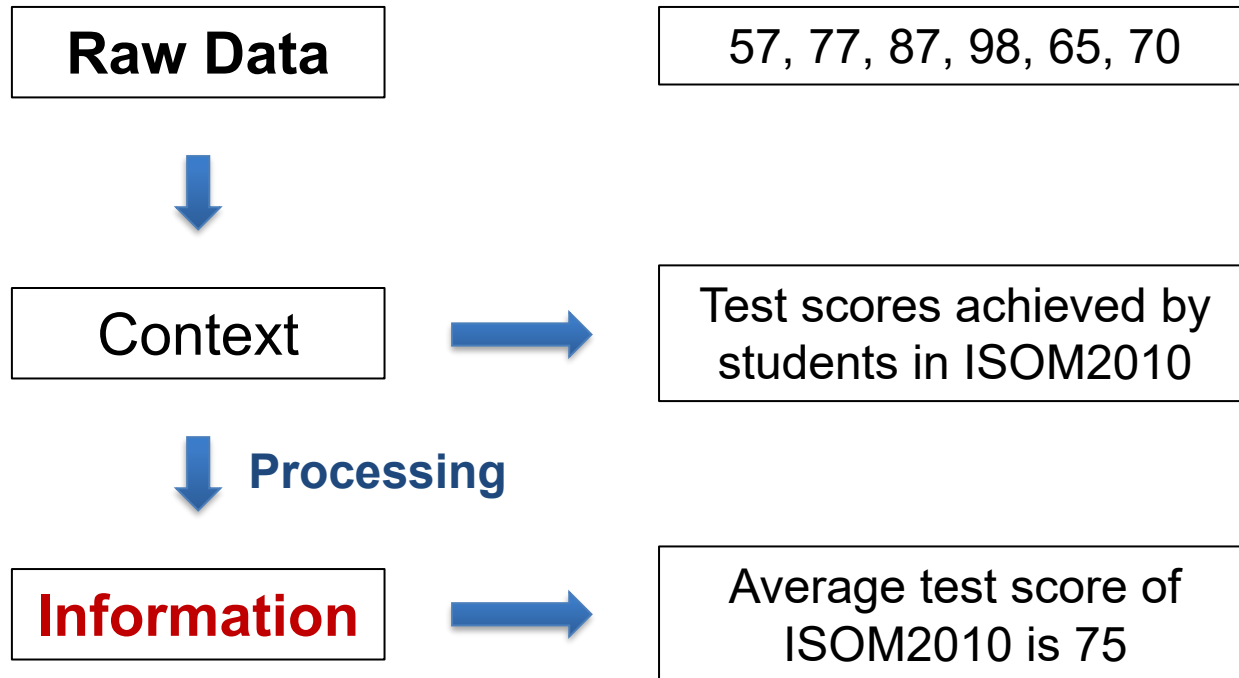
➤ *Transforming Data into Information*

- Identify the purpose of the data
- Organize the data
 - Select (or filter)
 - Sort
 - Summarize (or group)
 - Correct (if needed)
- Perform calculations
 - Simple (totals, averages, etc.)
 - Complex (e.g. statistical analyses)
- Format and present information



DATA, INFORMATION, KNOWLEDGE

▣ *Data & Information - Example*



DATA, INFORMATION, KNOWLEDGE

Data & Information - Example

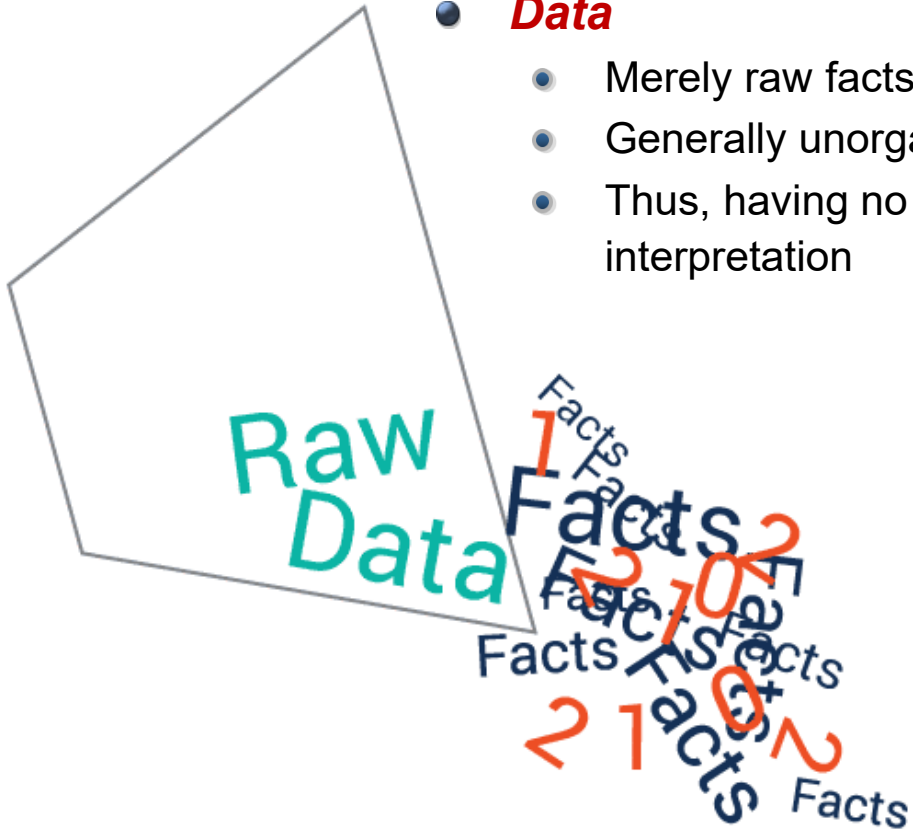
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1	cust_id	membership_level	order_number	order_date	order_time_range	delivery_date	delivery_timeslot	app_type	device_type	sales_application	payment_gateway	card_type	area	district	delivery_district	order_value
2	bX4Ac3X8IJ	NORMAL	dZZ9FHN0Dw4f	2021-01-08	00-09	2020-01-10	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_VISA		NT	YUEN L	NT-TM	450 - 499
3	pZssRbRxcL	NORMAL	M1vl+tpfZH8Kx	2021-01-08	15-12	2020-01-10	14:00 - 18:00	MAIN	Web	Web	payment_gateway_VISA		HK	NORTH	ETH	1000 - 1099
4	dP0lj0Cz12	NORMAL	SQJccer5VHZW	2021-01-08	00-09	2020-01-09	18:00 - 21:00	MAIN	Web	Web	payment_gateway_VISA		NT	TSEUNC	ETK-ZFA	150 - 199
5	sbQyaP+X1c	VIP	M2FCMzImPjJiz	2021-01-08	12-9	2020-01-10	10:00 - 14:00	MAIN	iOS	WebMobile	payment_gateway_VISA		NT	MA ON	NT-TSM	850 - 899
6	tA9xTfi3Znx	NORMAL	Lk8N/nbc67+M	2021-01-08	15-12	2020-01-11	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_VISA		NT	TSEUNC	ETK	1500 - 1599
7	yE5F2pO4N	NORMAL	qNspXMB8+RyL	2021-01-08	00-09	2020-01-09	14:00 - 18:00	MAIN	Android	WebMobile	payment_gateway_VISA		KLN	KOWLO	WTK-KLS	350 - 399
8	fDK03mIXb	NORMAL	qRtPDnLx2V7B\	2021-01-08	15-12	2020-01-10	10:00 - 14:00	MAIN	Web	Web	payment_gateway_VISA		KLN	TSIM S	WTK	450 - 499
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10	Dv0T4JL2o+	VIP	EKb/56ArDK9H	2021-01-08	12-9	2020-01-10	18:00 - 21:00	MAIN	Web	Web	payment_gateway_Master		KLN	TO KW/	ETX	950 - 999
11	5PUofJ8pT+	VIP	3yCu8ofWUZR	2021-01-08	21-24	2020-01-10	14:00 - 18:00	MAIN	Android	WebMobile	payment_gateway_Master		KLN	HUNG F	WTK-HOM	600 - 649
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14	y1fKzZuptQ	VIP	si21B1hqYv56k	2021-01-08	12-9	2020-01-11	14:00 - 18:00	MAIN	Android	WebMobile	payment_gateway_VISA		HK	KENNEI	WTH	1200 - 1299
15	AdvkofNhef	NORMAL	D7oGh+UGWJY	2021-01-08	15-18	2020-01-10	14:00 - 18:00	MAIN	Web	Web	payment_gateway_Master		NT	TSUEN	NT-RGS	100 - 149
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17	OmtO CtjnM	VIP	A67z6ks/xvqkP	2021-01-08	12-9	2020-01-11	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_Master		NT	TIN SHI	NT-TM	250 - 299
18	4t2sJl2395n	NORMAL	jhtzLYs8Bbk6y	2021-01-08	00-09	2020-01-09	10:00 - 14:00	MAIN	iOS	WebMobile	payment_gateway_OmmGX58		KLN	TAI KO	WTK	800 - 849
19	pk15BdS147	GOLDVIP	ECOPallIMTx0H	2021-01-08	00-09	2020-01-09	10:00 - 14:00	MAIN	iOS	WebMobile	payment_gateway_VISA		KLN	TSIM S	WTK	4000 - 4099
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21	qRoAXtzeuf	VIP	pMr1FuR2d0cF	2021-01-08	00-09	2020-01-09	18:00 - 21:00	MAIN	Android	WebMobile	payment_gateway_Master		NT	MA ON	NT-SSC	150 - 199
22	Dv0T4JL2o+	VIP	EKb/56ArDK9H	2021-01-08	12-9	2020-01-10	18:00 - 21:00	MAIN	Web	Web	payment_gateway_Master		KLN	TO KW/	ETX	950 - 999
23	MrA9ME9v+	VIP	LHYwmgaNU8C	2021-01-08	00-09	2020-01-09	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_VISA		NT	TSUEN	NT-RGS	250 - 299
24	TAwxb+6r	NORMAL	7Y7hJnlI2IAMf	2021-01-08	15-18	2020-01-10	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_VISA		NT	TSING Y	NT-TIV	250 - 299
25	5406BpWGJ	VIP	IBSszZGmUCh+k	2021-01-08	18-21	2020-01-11	10:00 - 14:00	MAIN	Android	WebMobile	payment_gateway_VISA		NT	SAI KUN	ETK	1100 - 1199
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28	BfnT1HSXs+	VIP	JEBoV8tXrnFNk	2021-01-08	12-9	2020-01-10	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_VISA		KLN	MEI FO	WTK-MEI	300 - 349
29	mrdO42A73	GOLDVIP	TADefW99Ohv	2021-01-08	15-12	2020-01-11	10:00 - 14:00	MAIN	iOS	WebMobile	payment_gateway_VISA		HK	WAN CI	WTH	250 - 299
30	YA2UI28ySL	VIP	qbDOX/EefQuE	2021-01-08	21-24	2020-01-14	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_VISA		KLN	HUNG F	WTK-HOM	2900 - 2999
31	qRoAXtzeuf	VIP	pMr1FuR2d0cF	2021-01-08	00-09	2020-01-09	18:00 - 21:00	MAIN	Android	WebMobile	payment_gateway_Master		NT	MA ON	NT-SSC	150 - 199
32	qRoAXtzeuf	VIP	pMr1FuR2d0cF	2021-01-08	00-09	2020-01-09	18:00 - 21:00	MAIN	Android	WebMobile	payment_gateway_Master		NT	MA ON	NT-SSC	150 - 199
33	QCcutUPkT2	VIP	4hVzqtW/xsRM	2021-01-08	00-09	2020-01-11	14:00 - 18:00	MAIN	iOS	WebMobile	payment_gateway_Master		NT	LAI KIN	WTK	250 - 299
34	EOHo7h1AZ	NORMAL	TnW6xpW4nFB	2021-01-08	18-21	2020-01-13	18:00 - 21:00	MAIN	iOS	WebMobile	payment_gateway_VISA		KLN	YAU TO	ETK-YTS	200 - 249
35	8oK+AJSIYr	VIP	g7FuRdVExjBxh	2021-01-08	00-09	2020-01-09	14:00 - 18:00	MAIN	Android	WebMobile	payment_gateway_VISA		KLN	LAM TI	ETK-SCE	250 - 299
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37	8oK+AJSIYr	VIP	g7FuRdVExjBxh	2021-01-08	00-09	2020-01-09	14:00 - 18:00	MAIN	Android	WebMobile	payment_gateway_VISA		KLN	LAM TI	ETK-SCE	250 - 299
38	Mr2qJYfiwV	VIP	izGn6GBBeKEqf	2021-01-08	21-24	2020-01-11	18:00 - 21:00	MAIN	Android	WebMobile	payment_gateway_Master		HK	AP LEI	C WTH	2500 - 2599
39	wMMzAg+n	GOLDVIP	xDTQvktGFg8O	2021-01-08	15-18	2020-01-11	14:00 - 18:00	MAIN	Android	WebMobile	payment_gateway_OmmGX58		NT	TSUEN	NT-TW	300 - 349
40	BV0Fh69I2Z	NORMAL	jQVFBBN+wEDf	2021-01-08	00-09	2020-01-09	18:00 - 21:00	MAIN	iOS	WebMobile	payment_gateway_Master		KLN	KOWLO	WTK	800 - 849

DATA, INFORMATION, KNOWLEDGE

▣ *Data*

- **Data**

- Merely raw facts/observations
- Generally unorganized/unprocessed
- Thus, having no value/meaning because of lack of context and interpretation



=

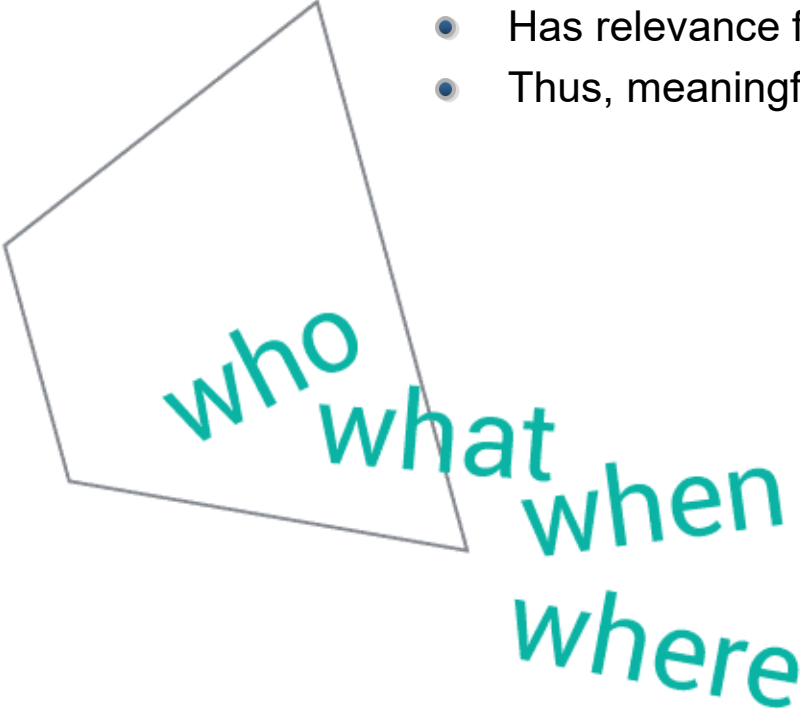
a collection of
facts in a raw or
unorganized form

Base building block - Raw **Data**

DATA, INFORMATION, KNOWLEDGE

Information

- **Information**
 - Data being organized or structured
 - Has relevance for a specific purpose or context
 - Thus, meaningful, valuable, useful, and relevant



who
what
when
where

=

easier to measure,
visualize and analyze
data for a specific purpose

Second building block - Derived **Information**

DATA, INFORMATION, KNOWLEDGE

▣ *Knowledge*

- **Knowledge**

- Information combined with understanding, skills, and experience
- The content of a thought in the individual's mind
- Tacit by nature



Third building block - Relevant **Knowledge**

DATA, INFORMATION, KNOWLEDGE

📌 *Same data, different information*

Raw Data: "12:30 PM, Mong Kok, 1,000 Orders of HK\$60 Solo Meals"

Perspective (Context)	Extracted Information (Useful/Meaningful Data)	Business Action
Marketing	"Solo diners in Mong Kok are highly responsive to the 'One-person Canteen' campaign during lunch hours."	Send targeted push notifications to users in Mong Kok at 12:15 PM tomorrow.
Operations	"There is a severe shortage of riders in Mong Kok at 12:30 PM to handle high-volume, low-value orders."	Temporarily increase 'On-time Bonuses' for riders moving into the Mong Kok area.
Finance	"The current promotion is attracting high volume, but the thin profit margin per HK\$60 order is increasing our burn rate."	Re-evaluate the subsidy amount per order to ensure long-term profitability.

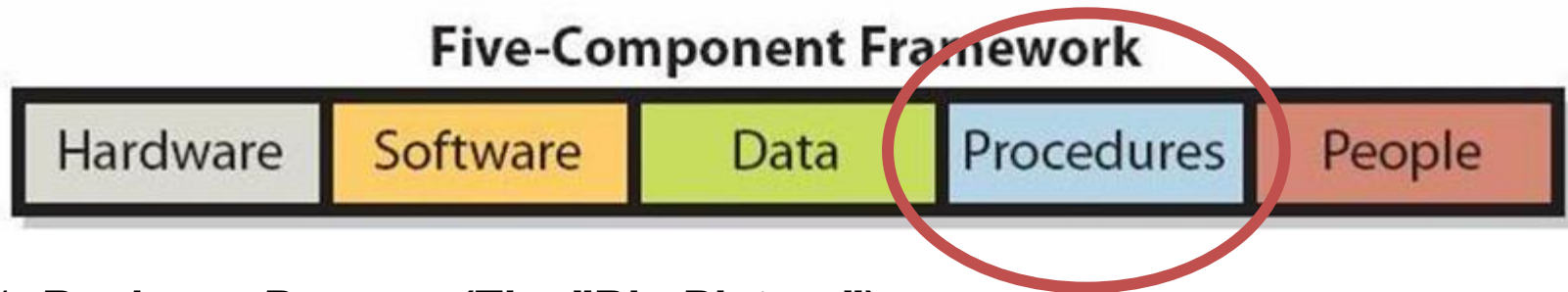
BUSINESS ACTION: practical application of Knowledge

📌 *Data → Information → Knowledge → Business Action*

Raw Data: "12:30 PM, Mong Kok, 1,000 Orders of HK\$60 Solo Meals"

- In the context of Information Systems, **Knowledge** is often defined as the ability to use information to make effective decisions or take appropriate actions.
- **Data:** "1,000 Solo Meals ordered in Mong Kok at 12:30 PM". (just a raw fact).
- **Information:** "There is a massive peak in demand from solo diners, but we are short on riders in that specific area". (This is data with context).
- **Knowledge:** "Based on past experience and current goals, we know that if we don't increase rider bonuses now, delivery times will exceed 30 minutes and we will lose customers". (This is information applied to a purpose).
- **Business Action:** Actually triggering the "On-time Bonus" in the app for riders in Mong Kok.

THE FIVE-COMPONENT FRAMEWORK OF IS



1. Business Process (The "Big Picture")

- **Definition:** A collection of logically related tasks and behaviors that an organization performs to accomplish a specific business goal.
- **Focus:** "The WHAT" — What is the overall objective?
- E.g., Order Fulfillment Process. This is the end-to-end journey from the moment a customer gets hungry to the moment they receive their meal.

2. Procedure (The "Instruction Manual")

- **Definition:** A specific set of rules or steps that guide individuals in how to perform their specific tasks within a process.
- **Focus:** "The HOW" — How should an actor use the system to complete a step?
- E.g., Proof-of-Delivery Procedure. The specific rule that a rider must take a photo of the food at the doorstep and "Slide to Complete" in the app.

THE FIVE-COMPONENT FRAMEWORK OF IS

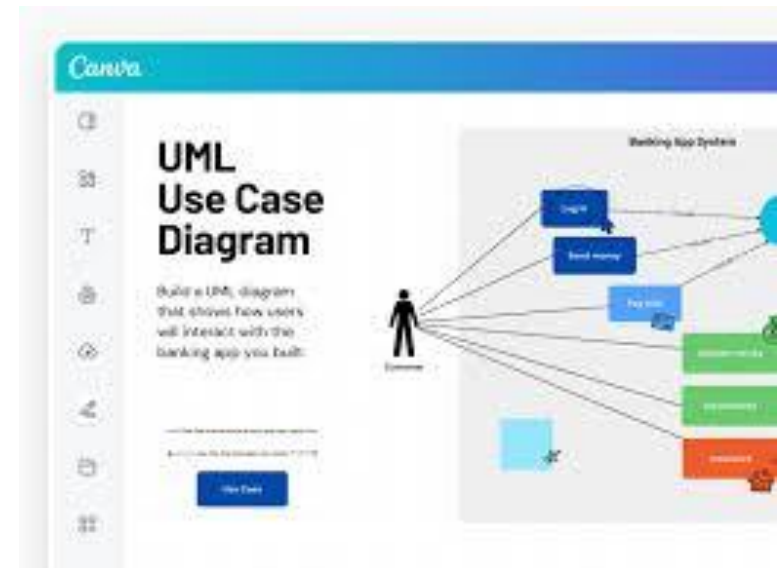
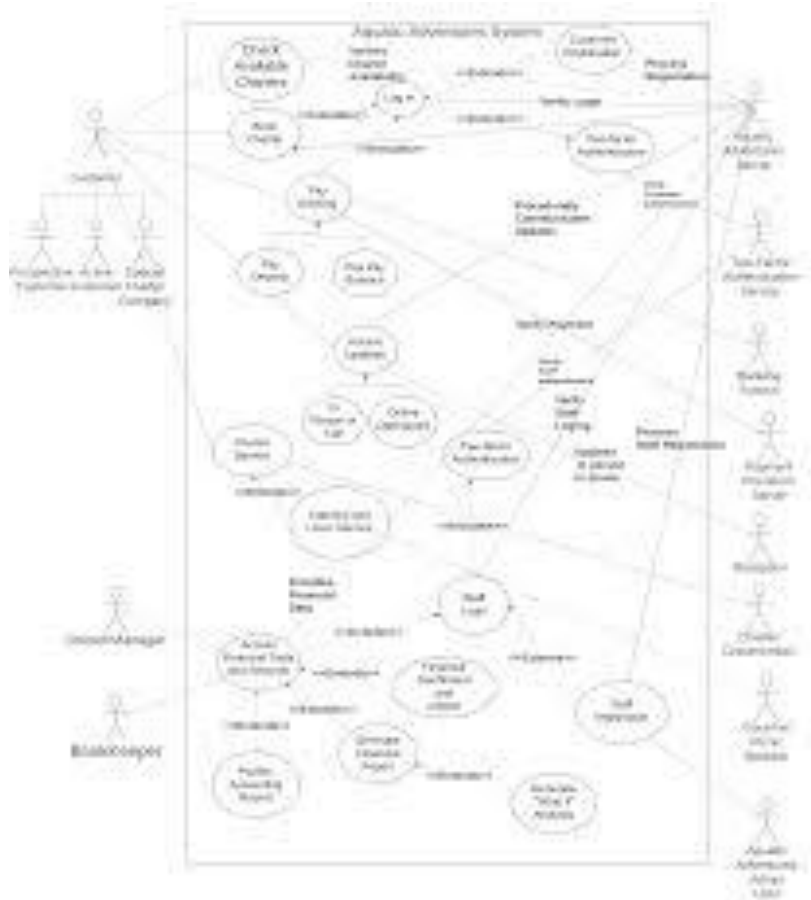


When an organization implements a new Information System, its **Procedures** must change.

Feature	Physical Bookstore	Online Bookstore	Agentic AI Era
Search & Discovery	Customer manually browses shelves or asks a clerk for help.	Customer inputs keywords into a search bar and filters results manually.	Customer states a goal (e.g., "Find a gift for a history buff") and AI identifies the best match.
Transaction Step	Customer brings a physical book to the counter and waits in line.	Customer adds to a digital cart and manually fills out payment/shipping forms.	Customer gives a single voice or text confirmation; AI executes payment and shipping autonomously.
Role of User	Walking, searching, and carrying.	Typing, clicking, and data entry.	Setting goals and approving recommendations.

THE FIVE-COMPONENT FRAMEWORK OF IS

Five-Component Framework



THE FIVE-COMPONENT FRAMEWORK OF IS

Five-Component Framework

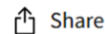


AI Adoption in the Workplace Still Low, 2 Years Later



Maria Rosala and Kate Moran

March 7, 2025



Summary: Designers may think AI features are now familiar to our users, but recent research suggests that adoption is still lower than we might think.

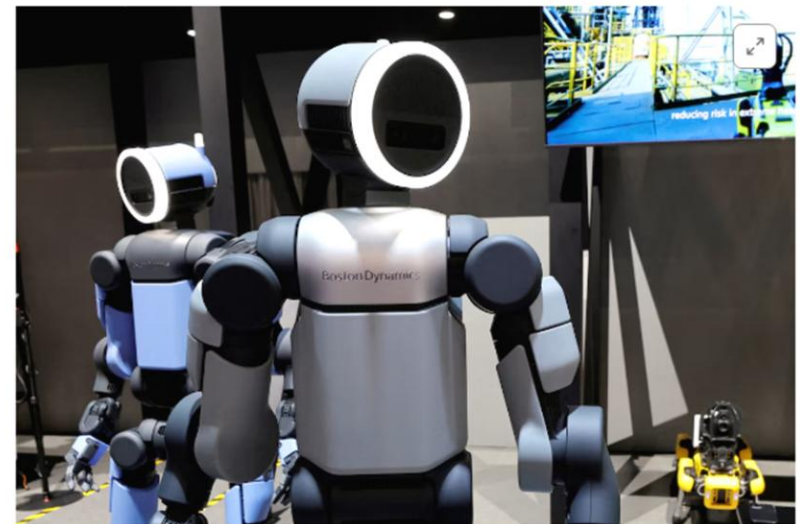


World ▾ Business ▾ Markets ▾ Sustainability ▾ Legal ▾ Commentary ▾ Technology ▾ Investigations ▾

Hyundai Motor's Korean union warns of humanoid robot plan, sees threat to jobs

By Heekyong Yang and Hyunjoo Jin

January 22, 2026 4:03 PM GMT+8 · Updated January 22, 2026



[1/2] Boston Dynamics Atlas robots are displayed in the Hyundai Motor Group booth during CES 2026, an annual consumer electronics trade show, in Las Vegas, Nevada, U.S. January 6, 2026. REUTERS/Steve Marcus/File Photo
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THE FIVE-COMPONENT FRAMEWORK OF IS



Most people think Information Systems is just about the first two boxes—Hardware and Software. **But as business leaders, your value is in the other three.**

Component	Difficulty to Change	Why?
Hardware	Easy	You can simply order new servers or tablets.
Software	Moderate	Installing new programs or AI models takes time but is predictable.
Data	Difficult	Cleaning and migrating massive amounts of data is a major project.
Procedures	Very Difficult	Changing long-standing business processes requires retraining everyone.
People	Most Difficult	Human habits, mindsets, and fears (like job loss to AI) are the hardest to shift.